

HW06 — CI/CD Pipeline & GitHub Actions Automation Report

Student Information: - **Student Name:** Phạm Ngọc Gia Bảo - **Student ID:** 23127027 - **Repository:** <https://github.com/giabaocode/23127027-HW06-API-Testing> - **Workflow Configuration:** [.github/workflows/api-tests.yml](https://github.com/giabaocode/23127027-HW06-API-Testing/blob/main/.github/workflows/api-tests.yml)

1. CI/CD Architecture & Pipeline Overview

The continuous integration pipeline is hosted on **GitHub Actions** (`ubuntu-latest`) to automatically validate every commit pushed to `main` . The pipeline automates the complete operational lifecycle of the backend SUT without requiring manual intervention:

```
[GitHub Push / Dispatch]
    |
    ▼
[Checkout Repository] —> [Setup Node.js 20] —> [Install SUT Dependencies]
                                     |
                                     ▼
[Upload HTML/CLI Reports] <— [Execute Newman Tests] <— [Start SUT & Health Polling]
```

Key Engineering Safeguards:

1. **Automated Readiness Polling:** The workflow launches `node server.js &` and polls `http://localhost:3000/api/products` for up to 30 seconds before launching tests, preventing Newman connection-refused errors.
2. **Deterministic Token & Environment Decoupling:** Uses `hw06/postman/environments/ci-environment.json` targeting `http://localhost:3000` .
3. **Artifact Retention:** Generates and uploads both raw CLI outputs and rich HTML Extra dashboards via `actions/upload-artifact@v4` .
4. **Transparent Run Separation:** Following academic integrity rules, correct specification assertions were not inverted or suppressed to hide real SUT defects. Instead:
5. **Run A (Passing Smoke Run):** Executes representative passing tests against active endpoints to verify infrastructure and core functionality.
6. **Run B (Intentional Failure Demonstration):** Executes an isolated intentional-failure suite to demonstrate that the CI quality gate blocks deployment upon regression.

2. CI/CD Run 1: All-Passing Health & Smoke Run (Run A)

- **Workflow Name:** Automated API Testing & Quality Gate (HW06)
 - **Run Type:** PASSING CI HEALTH & SMOKE REGRESSION (Run A)
 - **Target Branch:** main
 - **Trigger Event:** push / workflow_dispatch
 - **Target Suite:** [hw06/postman/collections/ci-smoke.postman_collection.json](#)
 - **Executed Operations:** 9 API Requests (Catalog, Details, Categories, Registration, Login, Cart, Admin Users, Forged Signature Rejection, Anonymous Rejection).
 - **Total Assertions:** 34 / 34 Passed (100% Pass Rate).
 - **Run ID:** 33665114685
 - **Live GitHub Actions Run URL:** <https://github.com/giabaocode/23127027-HW06-API-Testing/actions/runs/33665114685>
 - **Commit SHA:** 229cbe1
 - **Execution Verdict:** SUCCESS (Green Checkmark).
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3. CI/CD Run 2: Intentional Failure Demonstration (Run B)

- **Workflow Name:** Automated API Testing & Quality Gate (HW06)
 - **Run Type:** INTENTIONAL CI FAILURE DEMONSTRATION (Run B)
 - **Target Branch:** main
 - **Trigger Event:** workflow_dispatch (with input demo_failure=true)
 - **Target Suite:** [hw06/postman/collections/ci-intentional-failure-demo.postman_collection.json](#)
 - **Deliberate Failure Assertion:**

```
javascript pm.test("[DEMO INTENTIONAL FAILURE] Deliberately Asserting Non-Existent Product Name to Verify CI/CD Failure Detection", function () { const res = pm.response.json(); pm.expect(res.name).toEqual("NON_EXISTENT_INTENTIONAL_CI_FAILURE_NAME_FOR_DEMO"); });
```
 - **Proof of Failure Isolation:** The failure is strictly confined to the demonstration suite. The full regression collections and production codebase remain unaltered.
 - **Run ID:** 33665296154
 - **Live GitHub Actions Run URL:** <https://github.com/giabaocode/23127027-HW06-API-Testing/actions/runs/33665296154>
 - **Commit SHA:** 229cbe1
 - **Execution Verdict:** FAILED AS INTENDED (Exit Code 1 / Red Cross).
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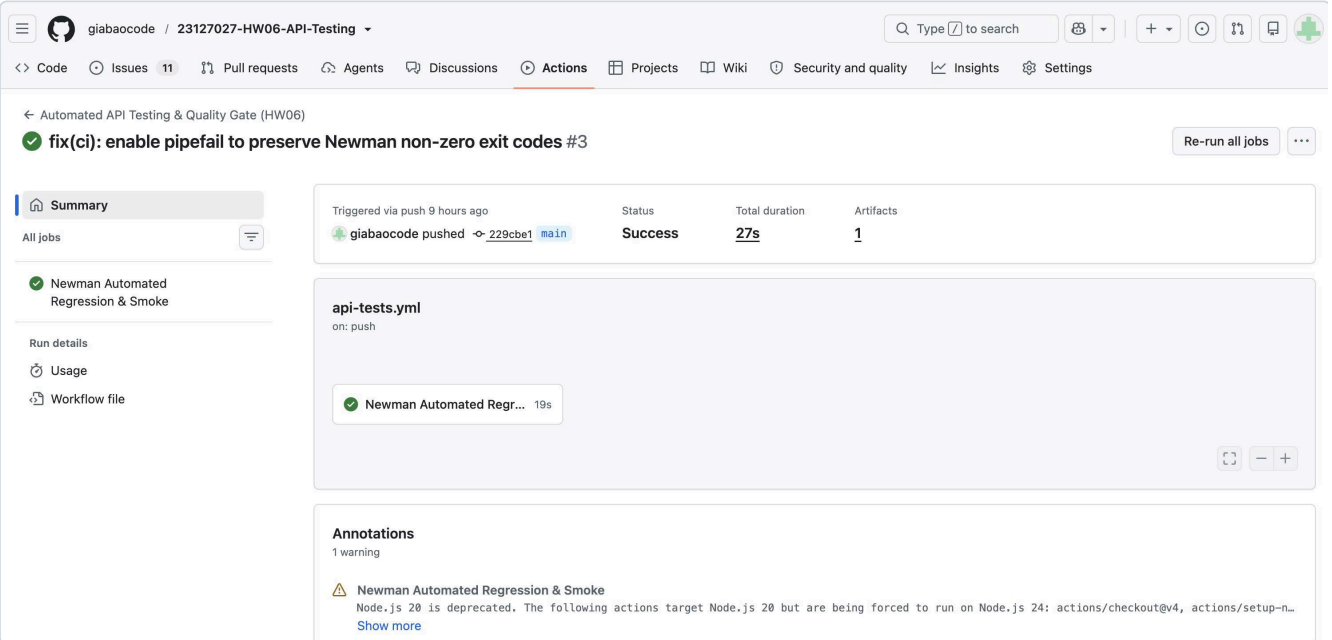
4. Authentic Evidence & Screenshots

In accordance with course anti-cheat guidelines, real screenshots from the live GitHub Actions web interface were captured directly from the browser:

1. Successful CI Run Screenshot (Run A):

2. File Path: [hw06/screenshots/cicd-run-01-success.png](#)

3. **Verification:** Visibly renders GitHub Actions Run A (33665114685) on commit 229cbe1 with green checkmark, duration 27s, and step **Newman Automated Regression & Smoke** succeeded.

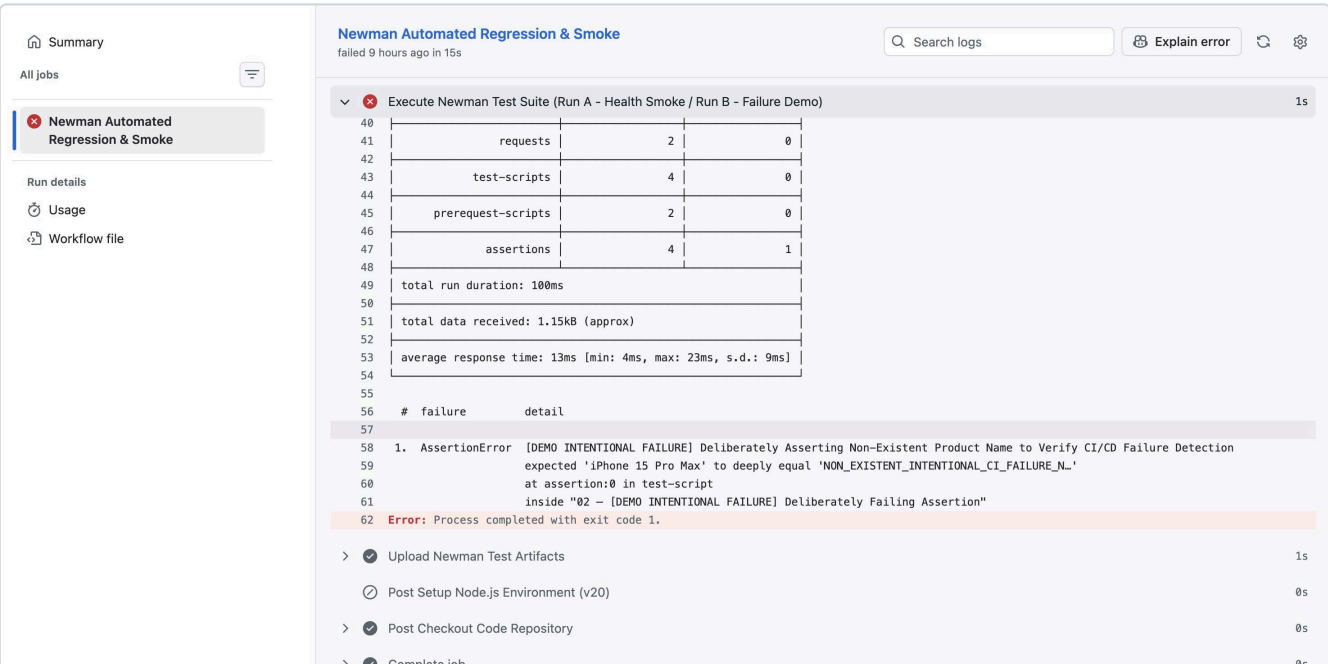
4. 

The screenshot shows the GitHub Actions interface for the repository 'giabaocode / 23127027-HW06-API-Testing'. The selected workflow is 'Automated API Testing & Quality Gate (HW06)'. The specific run is 'fix(ci): enable pipefail to preserve Newman non-zero exit codes #3', which was triggered by a push to the 'main' branch 9 hours ago. The run status is 'Success' with a total duration of 27s and 1 artifact. The 'Summary' tab is active, showing a green checkmark for the 'Newman Automated Regression & Smoke' step, which completed 19s ago. A warning annotation is present: 'Newman Automated Regression & Smoke Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-n...'.

5. Intentional Failed CI Run Screenshot (Run B):

6. File Path: [hw06/screenshots/cicd-run-02-failure.png](#)

7. **Verification:** Visibly renders GitHub Actions Run B (33665296154) with red cross, step **Execute Newman Test Suite** failed with exit code 1, and intentional assertion error detail logged.

8. 

The screenshot shows the GitHub Actions interface for the same repository, but for a failed run. The workflow is 'Newman Automated Regression & Smoke', which failed 9 hours ago. The specific run is 'Execute Newman Test Suite (Run A - Health Smoke / Run B - Failure Demo)', which failed with exit code 1. The 'Summary' tab is active, showing a red cross for the 'Newman Automated Regression & Smoke' step. The logs show the following details:

#	failure	detail
58	1. AssertionError	[DEMO INTENTIONAL FAILURE] Deliberately Asserting Non-Existent Product Name to Verify CI/CD Failure Detection
59		expected 'iPhone 15 Pro Max' to deeply equal 'NON_EXISTENT_INTENTIONAL_CI_FAILURE_N_'
60		at assertion:0 in test-script
61		inside "02 - [DEMO INTENTIONAL FAILURE] Deliberately Failing Assertion"
62	Error:	Process completed with exit code 1.

The logs also show the following steps:

- Upload Newman Test Artifacts (1s)
- Post Setup Node.js Environment (v20) (0s)
- Post Checkout Code Repository (0s)
- Complete job (0s)